Student Name: _____

Class Name: **ALEKS Integrated Mathematics II**

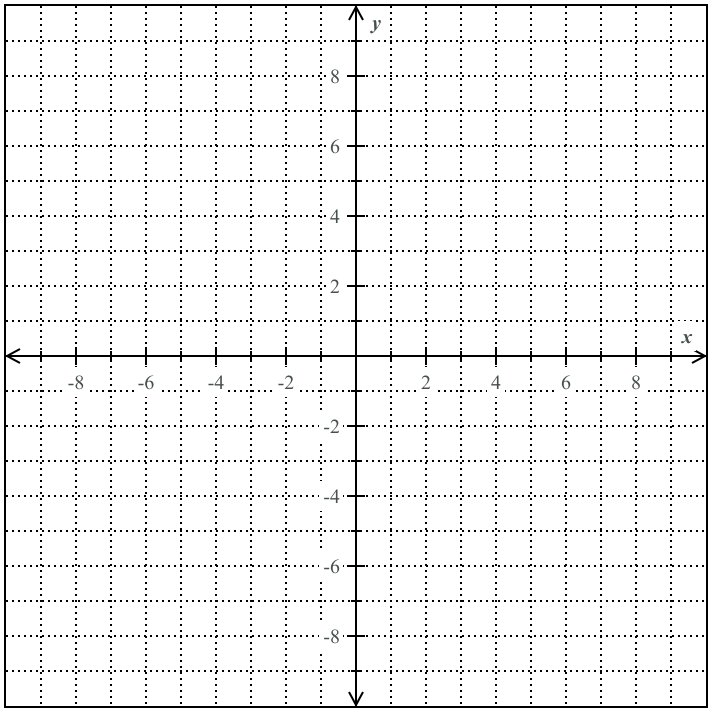
Number of Questions: **38**

Instructor Name: **Bodanske, Daniel**

Question 1 of 38

Graph the line.

$$y = 2x - 5$$



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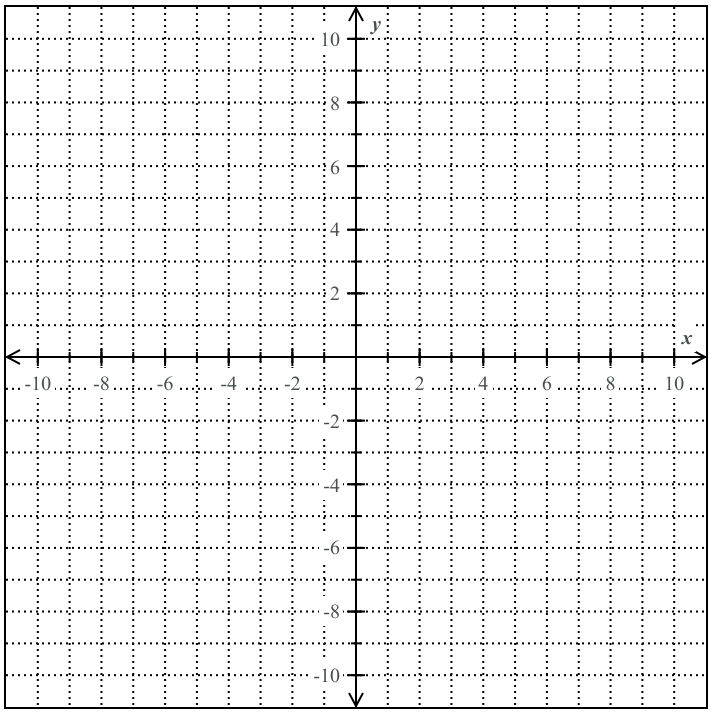
Question 2 of 38

Graph the line.

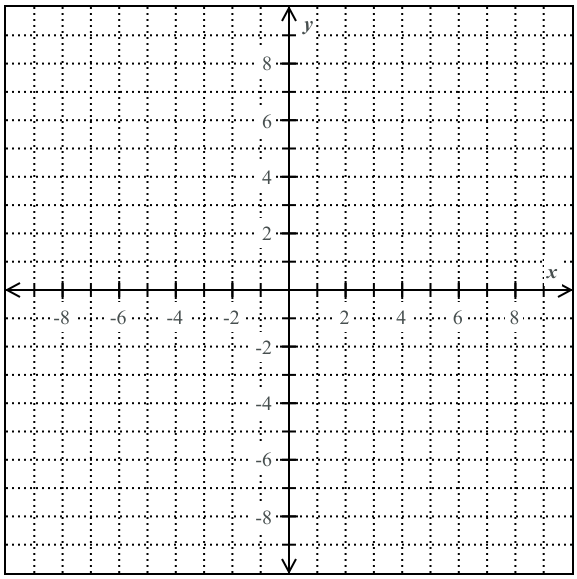
$$y = \frac{4}{3}x + 1$$



Question 3 of 38

Graph the line.

$$-3x + y = 3$$



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Question 4 of 38

Find the y-intercept and x-intercept of the line.

$$2x - y = 6$$

y-intercept: _____

x-intercept: _____

Question 5 of 38

The equation of a line is given below.

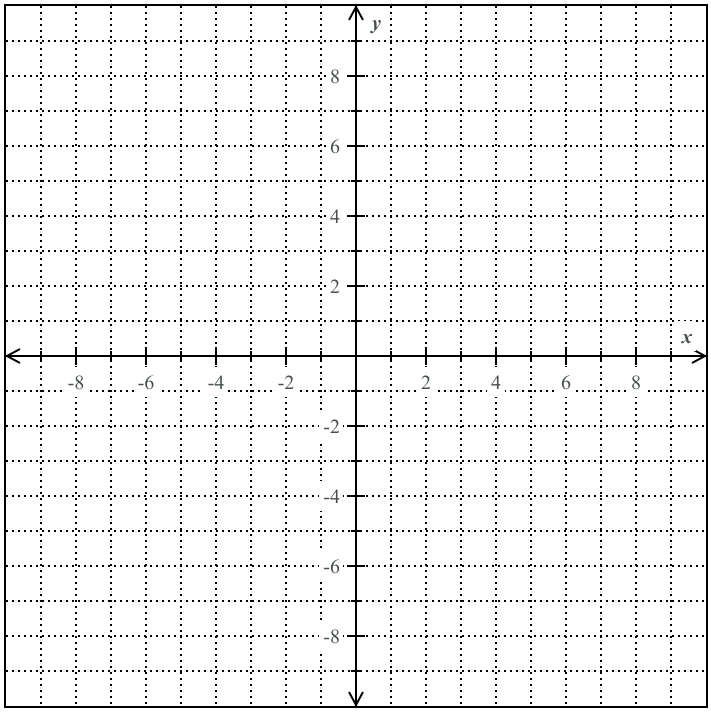
$$6x + 3y = -12$$

(a) Find the x-intercept and the y-intercept.

x-intercept: _____

y-intercept: _____

(b) Use the intercepts to graph the line.



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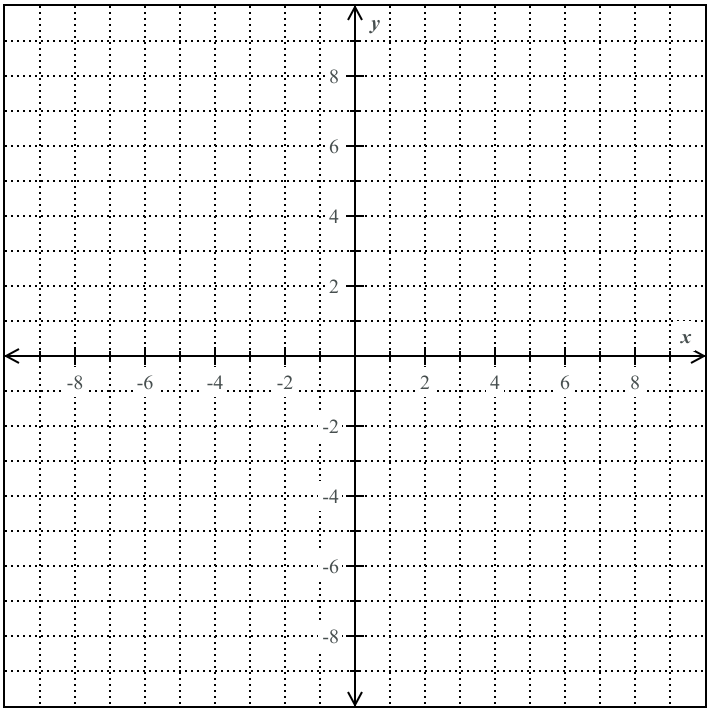
Question 6 of 38

Graph the line with slope $\frac{1}{2}$ and y-intercept 6.



Question 7 of 38

Find the slope and the y-intercept of the line.

$$-2x + 8y = -3$$

Write your answers in simplest form.

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Question 8 of 38

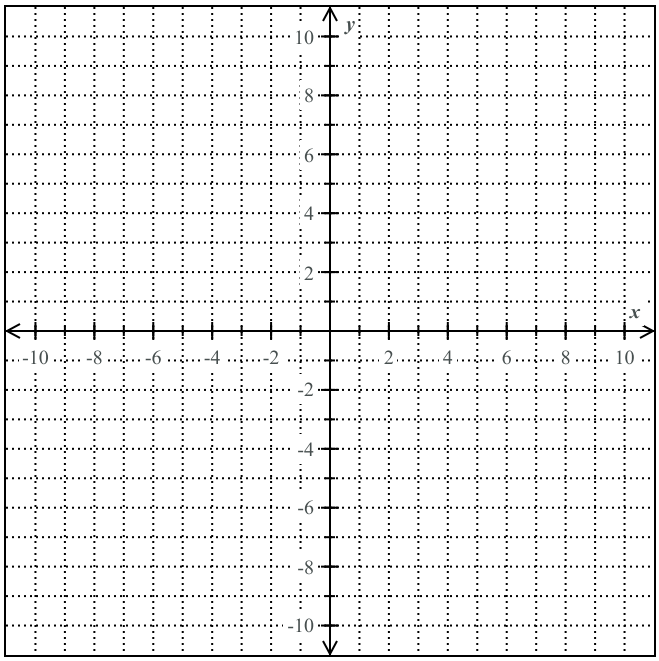
The equation of a line is given below.

$$-3x + 6y = 24$$

Find the slope and the y-intercept.
Then use them to graph the line.

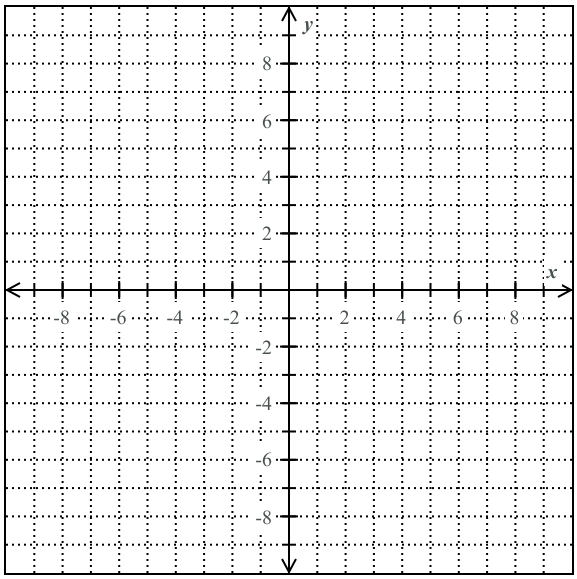
slope: _____

y-intercept: _____



Question 9 of 38

Graph the function $h(x) = 3x - 7$.



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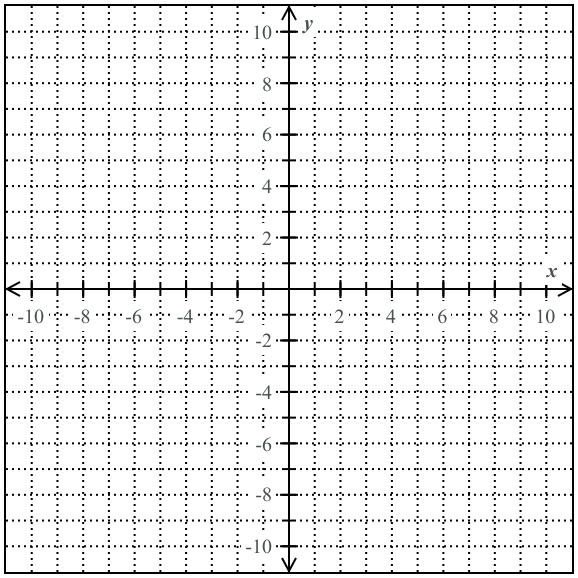
Question 10 of 38

Graph the function $f(x) = \frac{1}{3}x + 7$.



Question 11 of 38

For each ordered pair (x, y), determine whether it is a solution to the inequality $2x + 6y \geq -12$.

(x, y)	Is it a solution?	
	Yes	**No**
$(-6, 5)$	○	○
$(0, -2)$	○	○
$(4, -5)$	○	○
$(-8, -3)$	○	○

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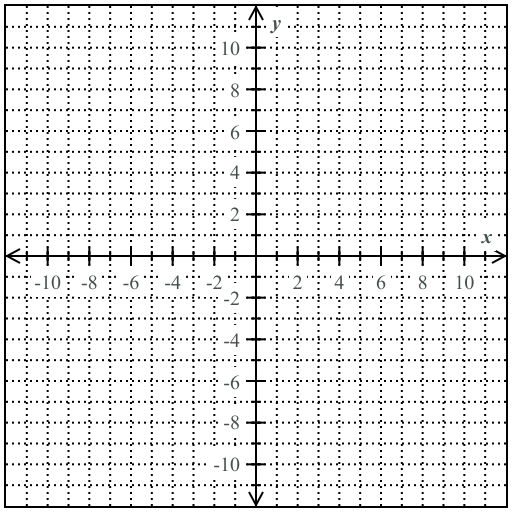
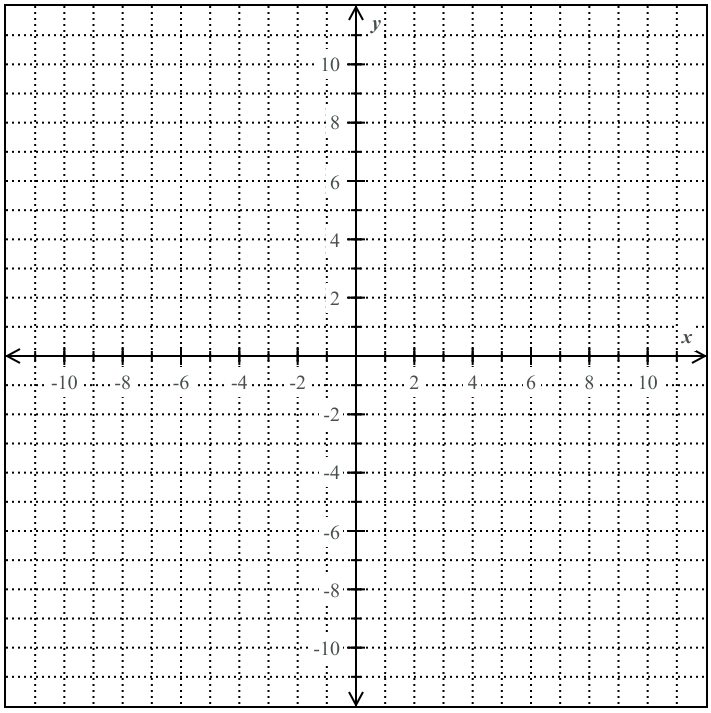
Question 12 of 38

Graph the inequality in the coordinate plane.

$$y > -3$$





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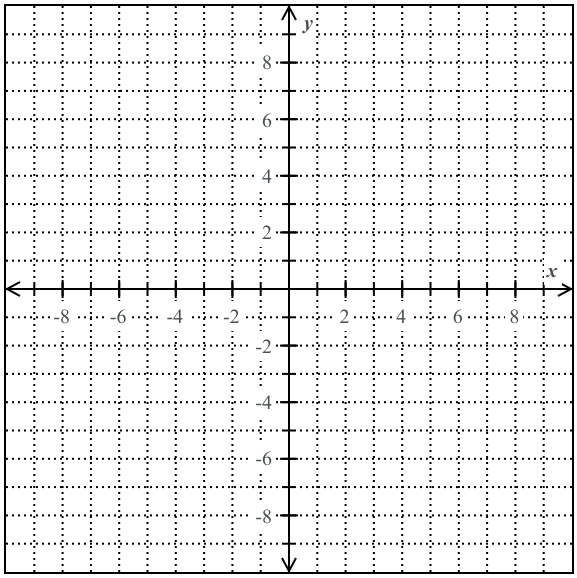
Question 13 of 38

Graph the inequality.

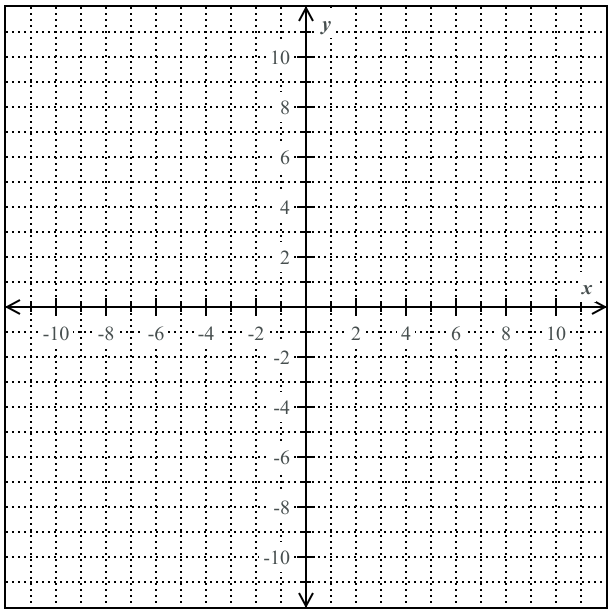
$$y \leq -2x - 5$$



Question 14 of 38

Graph the inequality.

$$-3x - 4y \geq -8$$



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Question 15 of 38

A small cargo plane currently has 135 pounds of cargo aboard. In addition, n boxes weighing 35 pounds each will be brought aboard.

Suppose that the total weight of the cargo on board must be at most p pounds. Using the values and variables given, write an inequality describing this.

Question 16 of 38

Suppose that the function f is defined, for all real numbers, as follows.

$$f(x) = \begin{cases} \frac{1}{2}x - 2 & \text{if } x \leq -2 \\ (x-1)^2 - 1 & \text{if } -2 < x < 2 \\ -2 & \text{if } x \geq 2 \end{cases}$$

Find $f(-2)$, $f(1)$, and $f(3)$.

$f(-2) =$ _____

$f(1) =$ _____

$f(3) =$ _____

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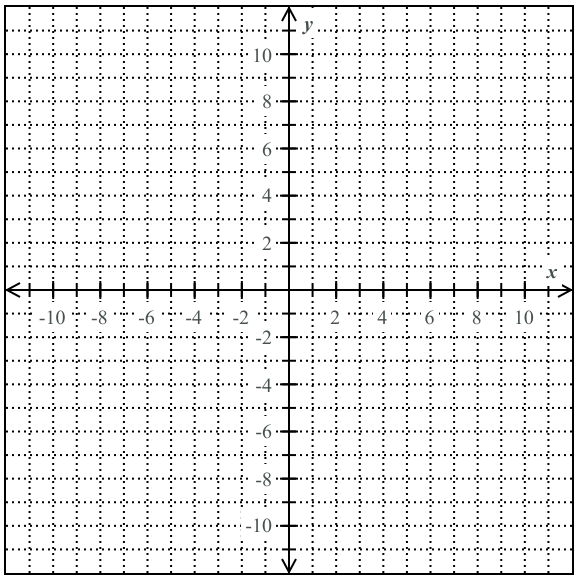
Question 17 of 38

Graph the equation.

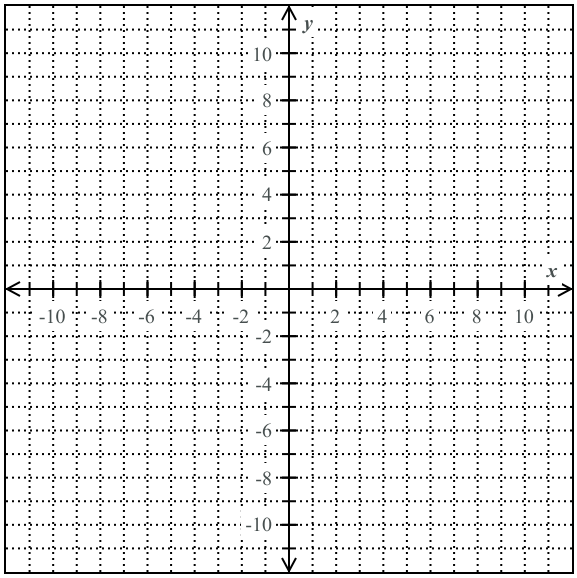
$$y = -|x|$$



Question 18 of 38

Graph the equation.

$$y = 4|x - 4|$$



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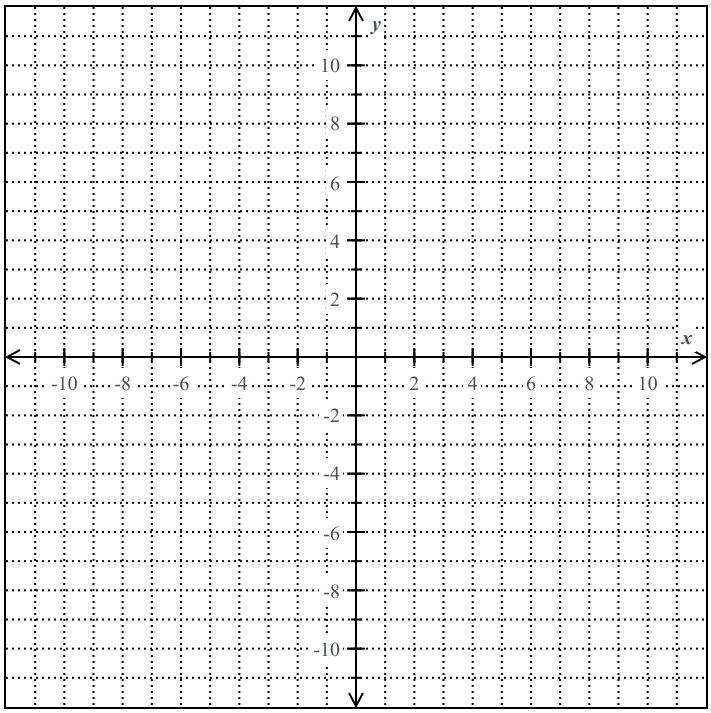
Question 19 of 38

Graph the equation.

$$y = 4\left|x - 3\right| + 4$$

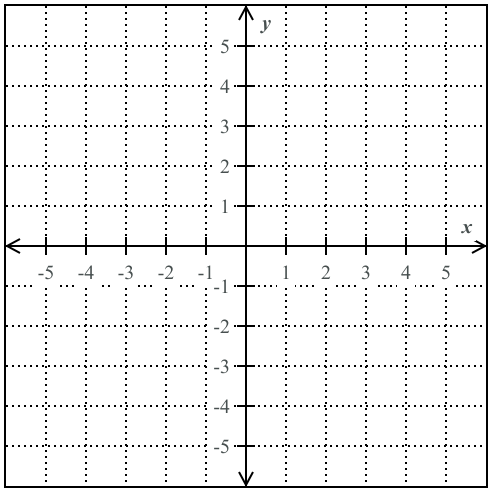


Question 20 of 38

Suppose that the function g is defined, for all real numbers, as follows.

$$g(x) = \begin{cases} -2 & \text{if } x \neq 0 \\ -5 & \text{if } x = 0 \end{cases}$$

Graph the function g.



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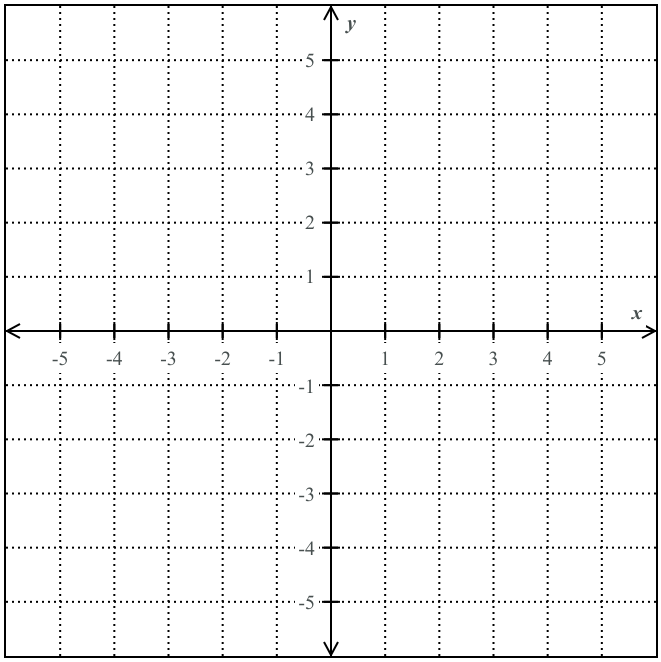
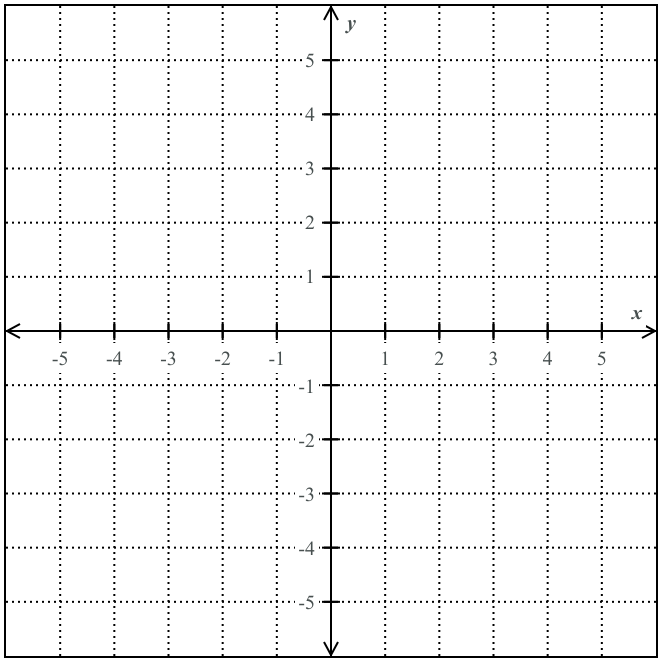
Question 21 of 38

Suppose that the function h is defined, for all real numbers, as follows.

$$h(x) = \begin{cases} -4 & \text{if } x \leq -2 \\ x-2 & \text{if } -2 < x \leq 3 \\ 3 & \text{if } x > 3 \end{cases}$$

Graph the function h.





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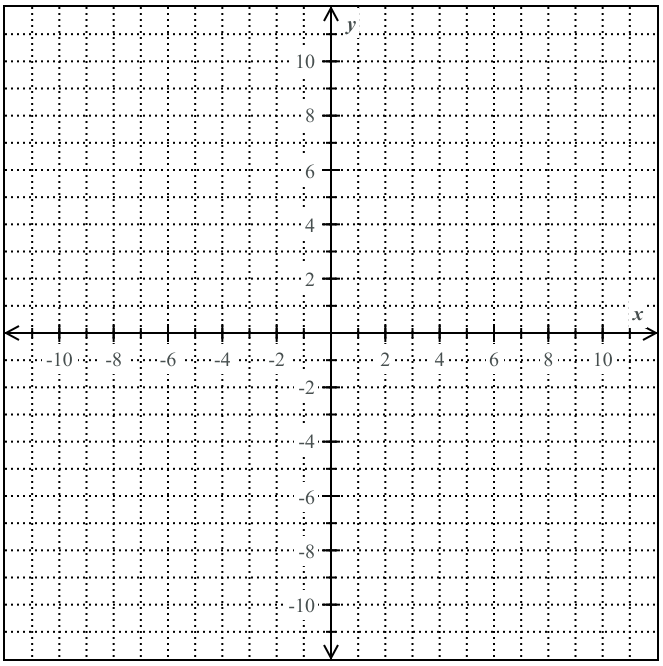
Question 22 of 38

Suppose that the function f is defined, for all real numbers, as follows.

$$f(x) = \begin{cases} x - 3 & \text{if } x \leq -2 \\ 4x + 5 & \text{if } x > -2 \end{cases}$$

Graph the function f. Then determine whether or not the function is continuous.



Is the function continuous?

- Yes
- No

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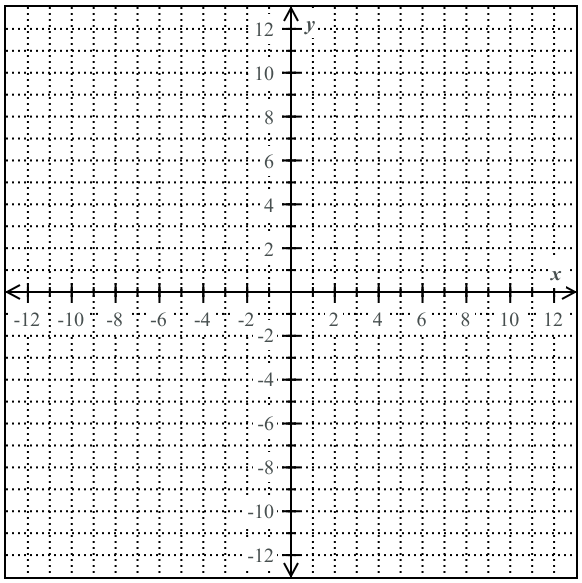
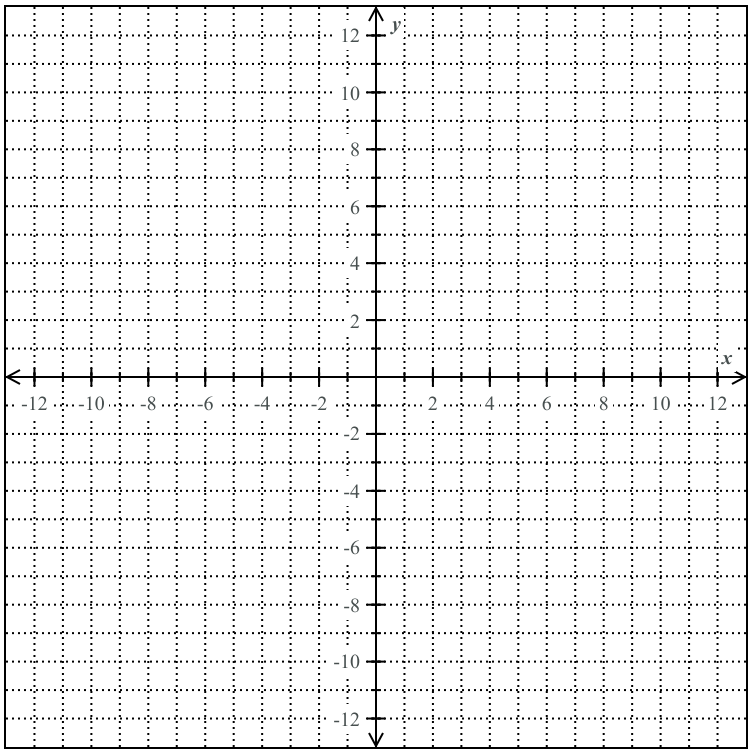
Question 23 of 38

Graph the function.

$$f(x) = 2x^2$$





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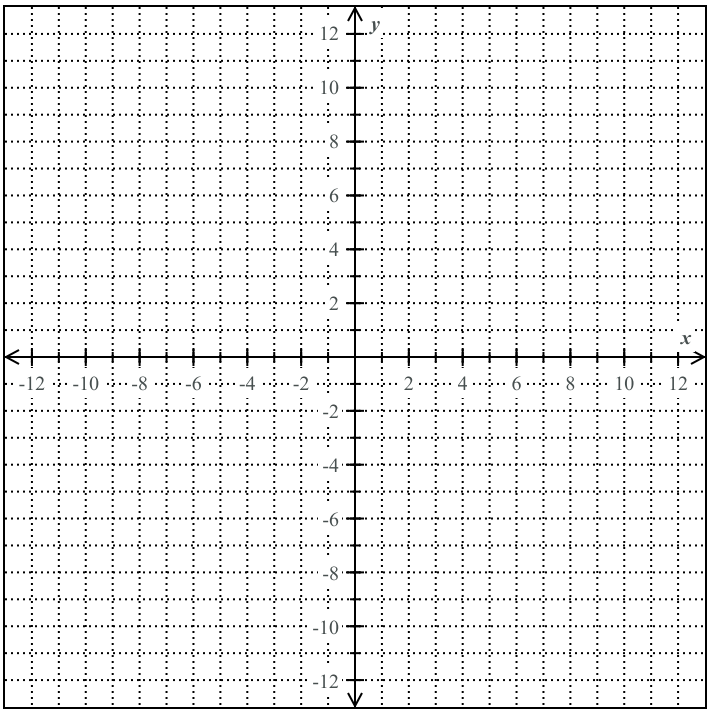
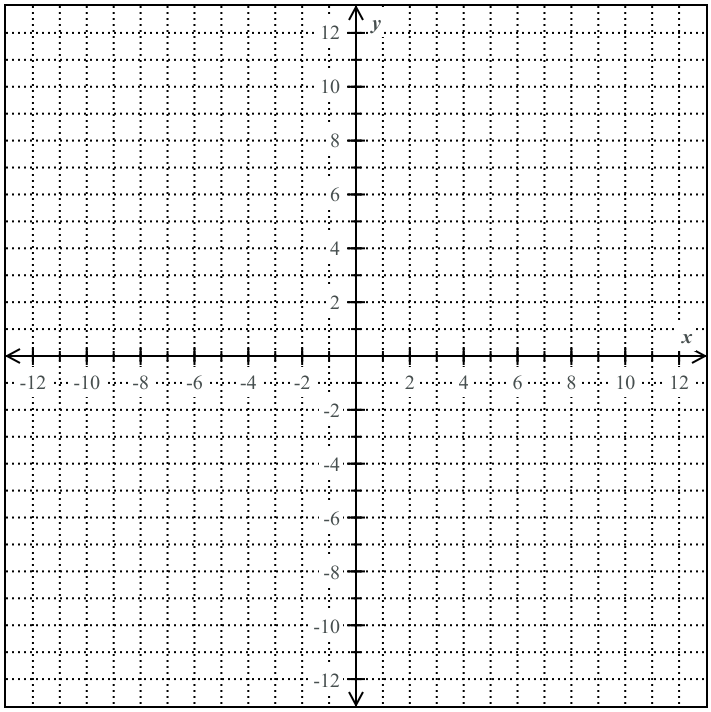
Question 24 of 38





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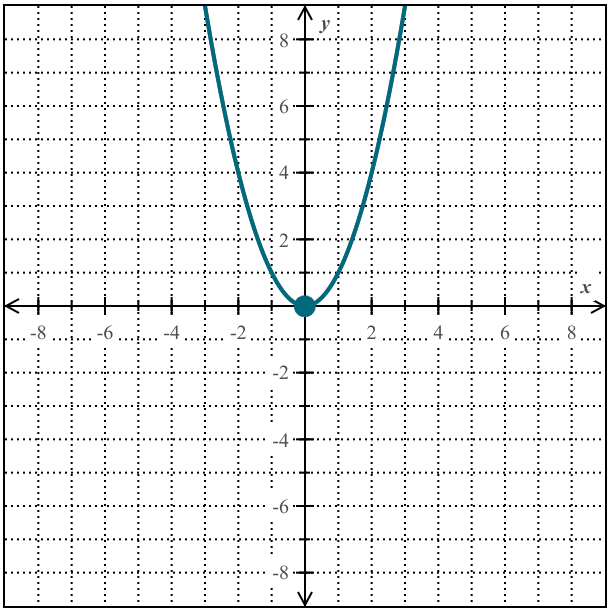
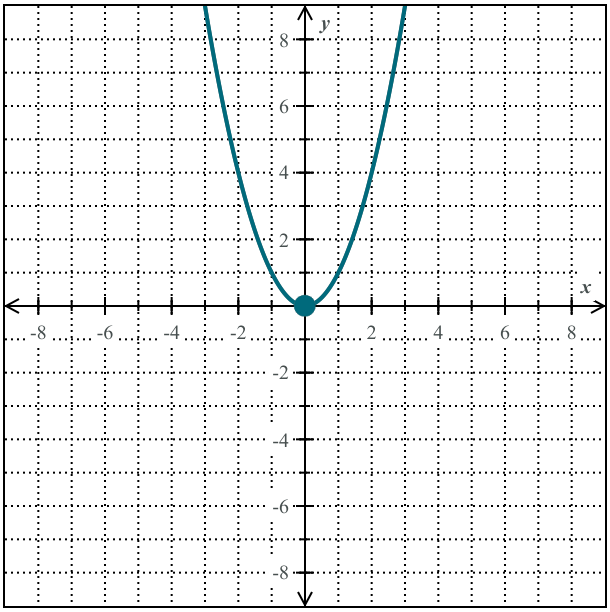
Question 25 of 38

Translate each graph as specified below.

(a) The graph of $y = x^2$ is shown. Translate it to get the graph of $y = (x-5)^2$.



(b) The graph of $y = x^2$ is shown. Translate it to get the graph of $y = x^2 + 3$.



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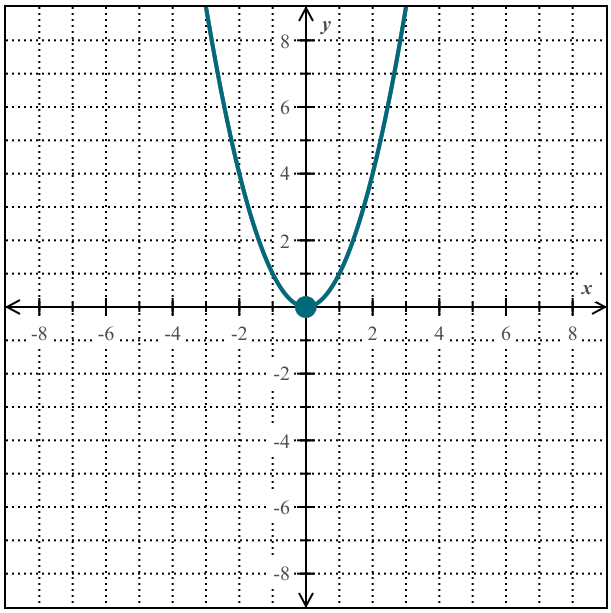
Question 26 of 38

Below is the graph of $y = x^2$.

Translate it to make it the graph of $y = (x+4)^2 + 2$.



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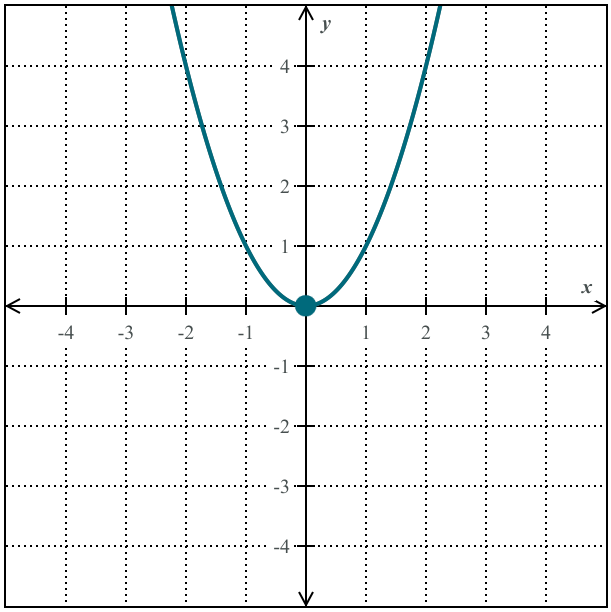
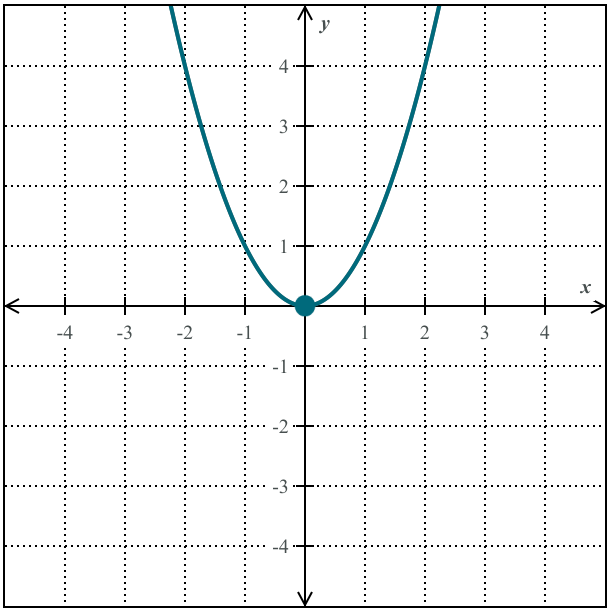
Question 27 of 38

The graph of $y=x^2$ is shown on each grid in parts (a) and (b) below.

(a) Use transformations to get the graph of $y=(-2x)^2$.



(b) Use transformations to get the graph of $y=-\frac{1}{2}x^2$.



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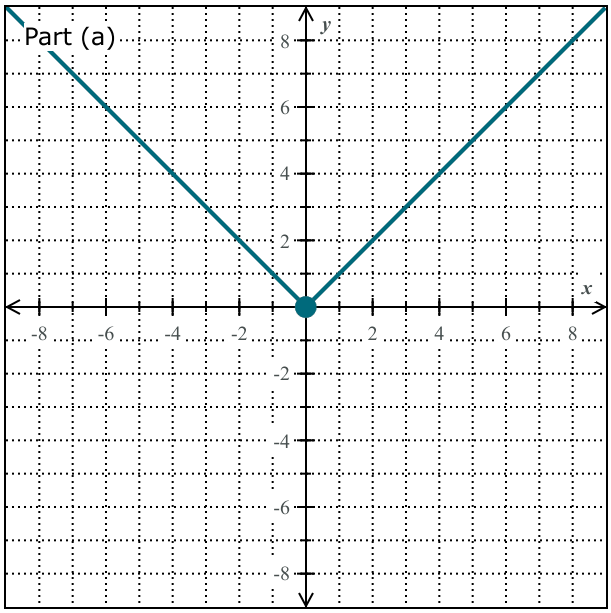
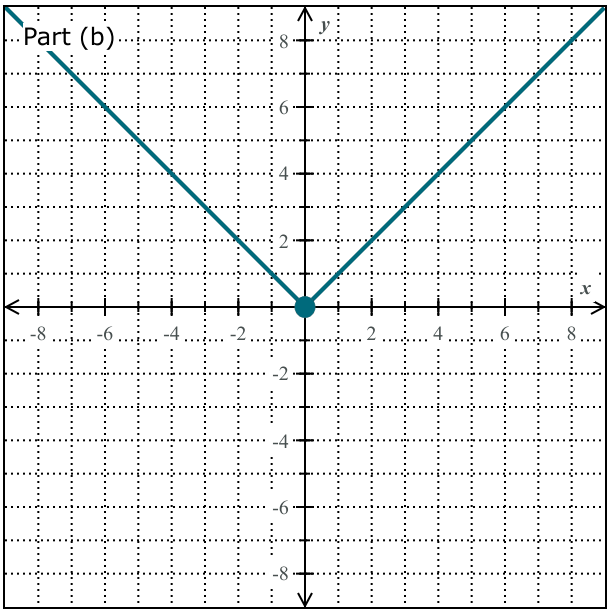
Question 28 of 38

Translate each graph as specified below.

(a) The graph of $y = |x|$ is shown. Translate it to get the graph of $y = |x - 1|$.

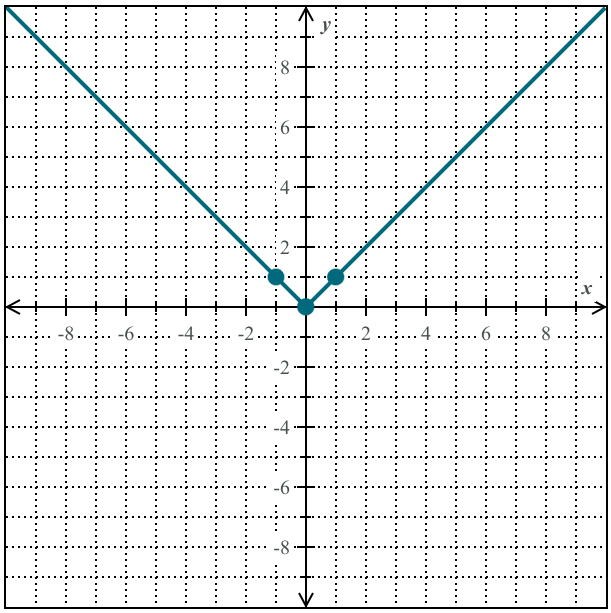
(b) The graph of $y = |x|$ is shown. Translate it to get the graph of $y = |x| + 5$.





Question 29 of 38

Below is the graph of $y = |x|$. Translate it to make it the graph of $y = |x - 3| - 4$.



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Question 30 of 38

The function g is defined as follows.

$$g(x) = 2x^2 - 9$$

If the graph of g is translated vertically upward by 7 units, it becomes the graph of a function f.

Find the expression for $f(x)$.

$f(x) =$ ☐

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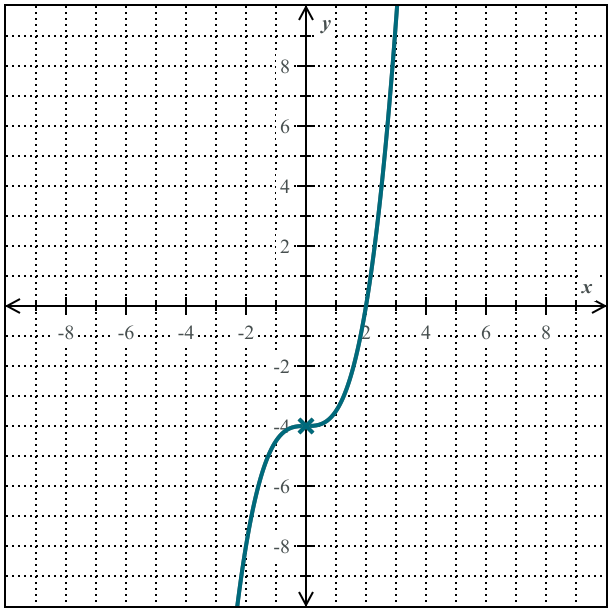
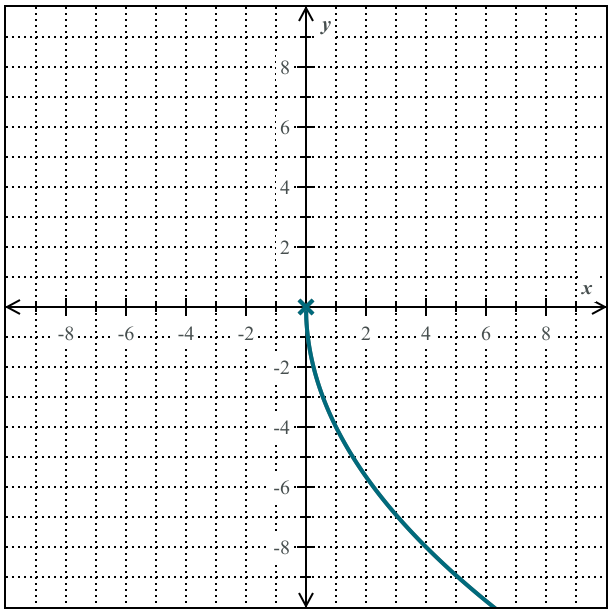
Question 31 of 38

Translate each graph as specified below.

(a) The graph of $y=f(x)$ is shown. Translate it to get the graph of $y=f(x+4)$.



(b) The graph of $y=g(x)$ is shown. Translate it to get the graph of $y=g(x)+3$.



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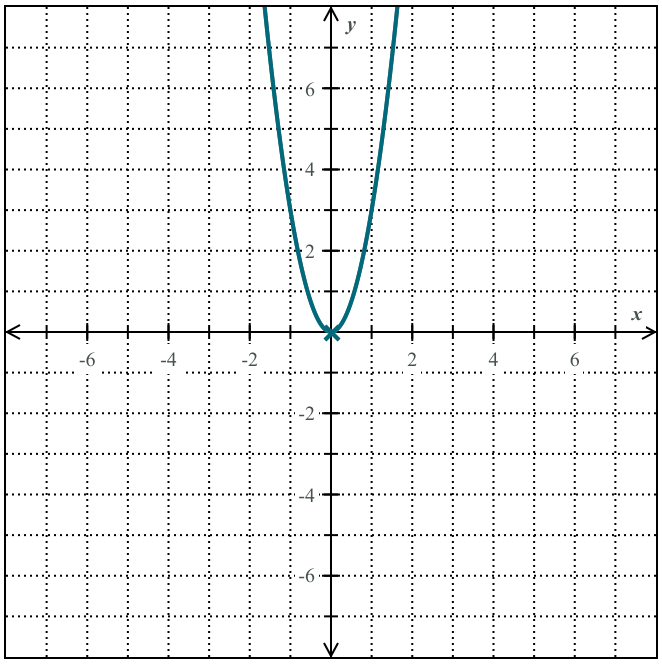
Question 32 of 38

Below is the graph of $y = 3x^2$.

Translate it to make it the graph of $y = 3(x-1)^2 - 2$.



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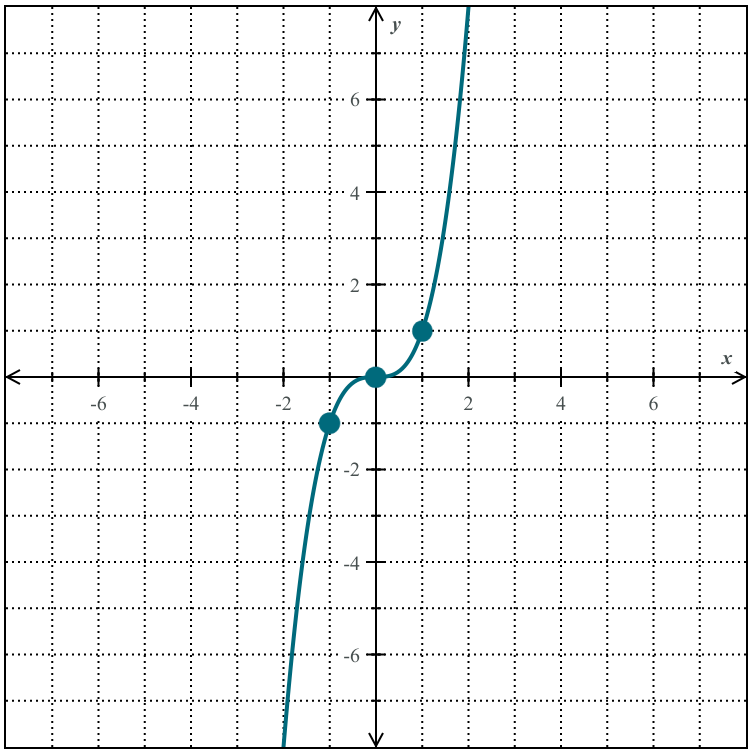
Question 33 of 38

Below is the graph of $y = x^3$.

Transform it to make it the graph of $y = 2(x-4)^3 + 3$.



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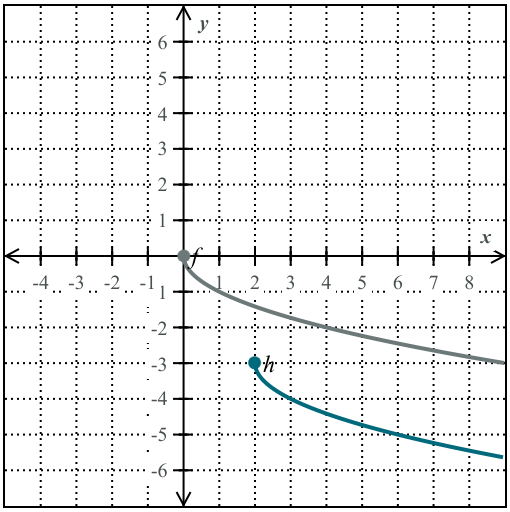
Question 34 of 38

The graph of f is translated a whole number of units horizontally and vertically to obtain the graph of h.

The function f is defined by $f(x) = -\sqrt{x}$.
Find the expression for $h(x)$.



$h(x) =$ _____

$h(x) =$ _____

Question 35 of 38

Find the y-intercept and the slope of the line.

$$5x - 3y = 3$$

Write your answers in simplest form.

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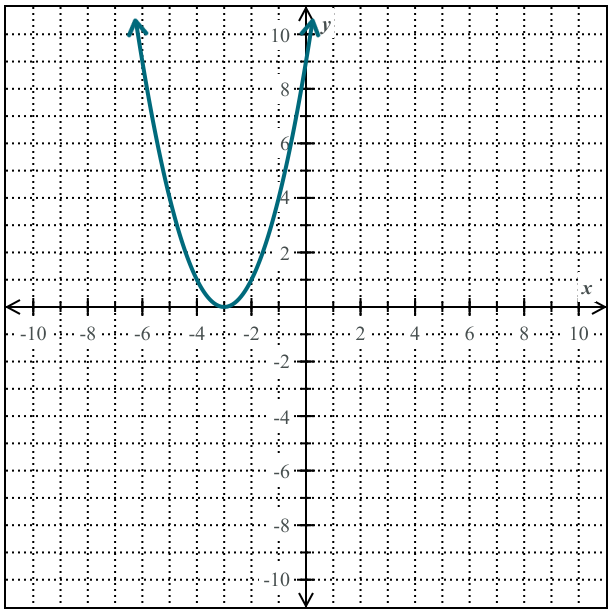
Question 36 of 38

The graph of a quadratic function with vertex $(-3,\ 0)$ is shown in the figure below. Find the range and the domain.



range: _____

domain: _____

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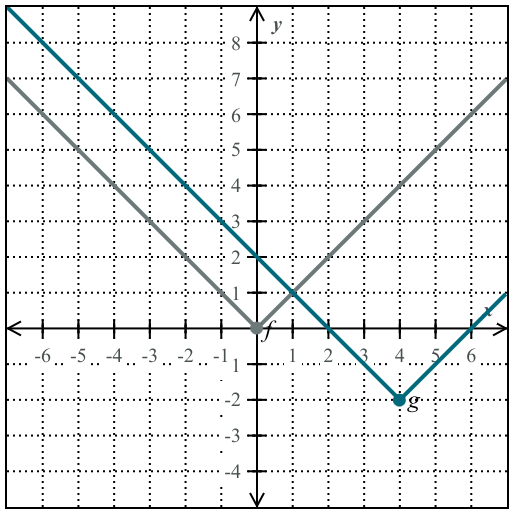
Question 37 of 38

The graph of f is translated a whole number of units horizontally and vertically to obtain the graph of g.

The function f is defined by $f(x) = |x|$.

Find the expression for $g(x)$.



$g(x) =$ _____

$g(x) =$ _____

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Question 38 of 38

For each function, state whether it is linear.

Function 1

$\{(5, 1), (6, 5), (7, 8), (8, 9)\}$

- Linear
- Not linear

Function 2

$\{(1, 0), (4, -2), (7, -4), (10, -2)\}$

- Linear
- Not linear

Function 3

x	y
-4	-5
0	-7
4	-9
8	-11

- Linear
- Not linear

Function 4

x	y
-3	3
-1	9
1	27
3	81

- Linear
- Not linear

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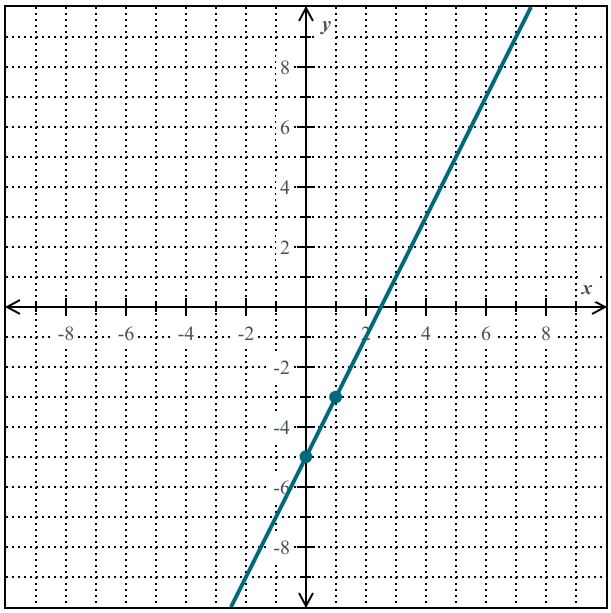
Class Name: **ALEKS Integrated Mathematics II** Number of Questions: **38**

Question 1 of 38



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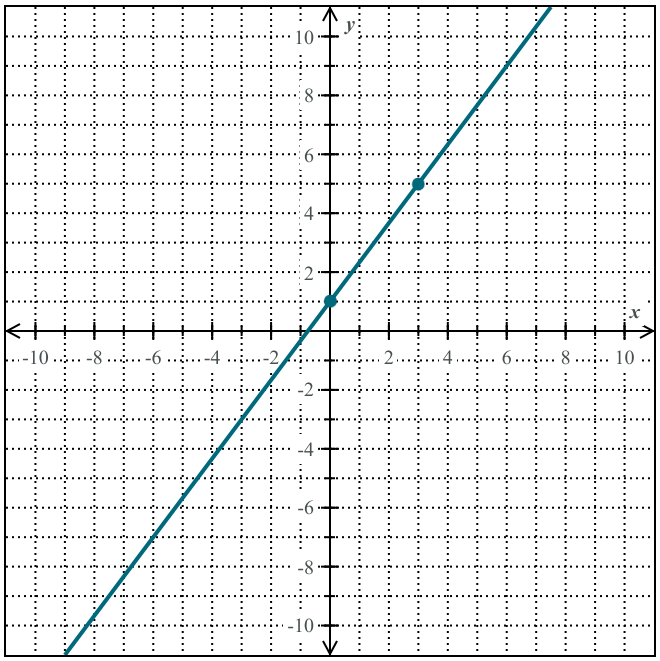

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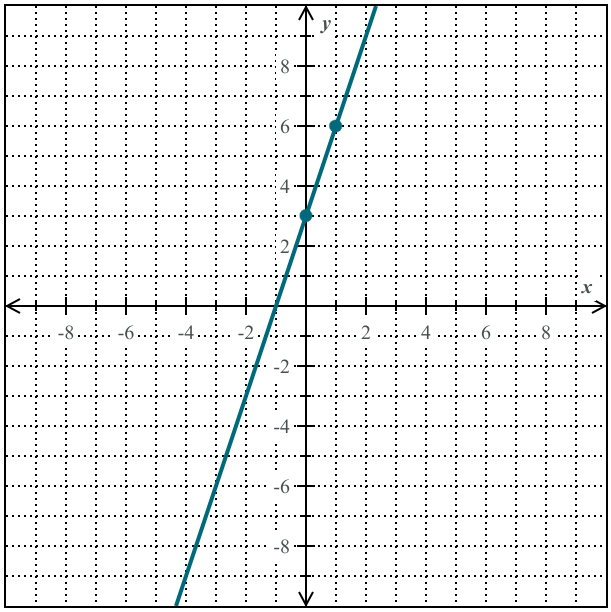


Question 4 of 38

y-intercept: -6

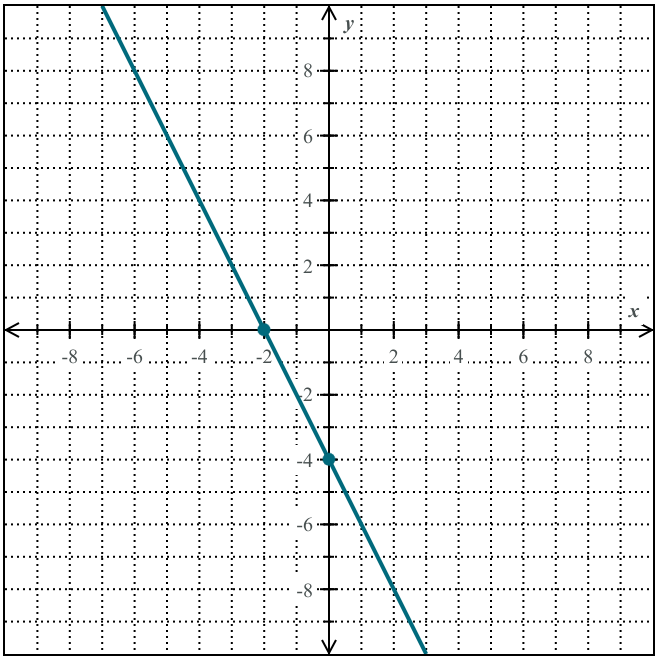
x-intercept: 3

Question 5 of 38

(a) x-intercept: -2

y-intercept: -4

(b)



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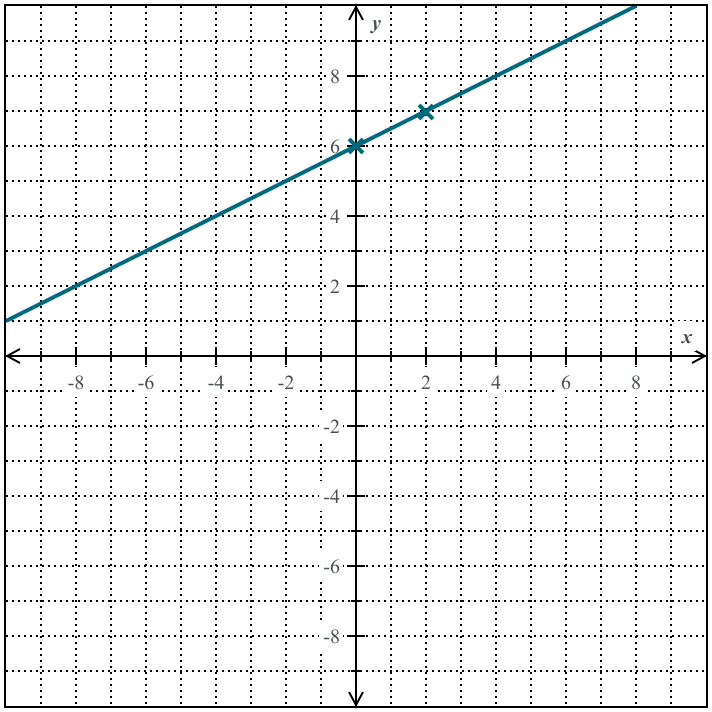

Question 7 of 38

slope: $\frac{1}{4}$

y-intercept: $-\frac{3}{8}$

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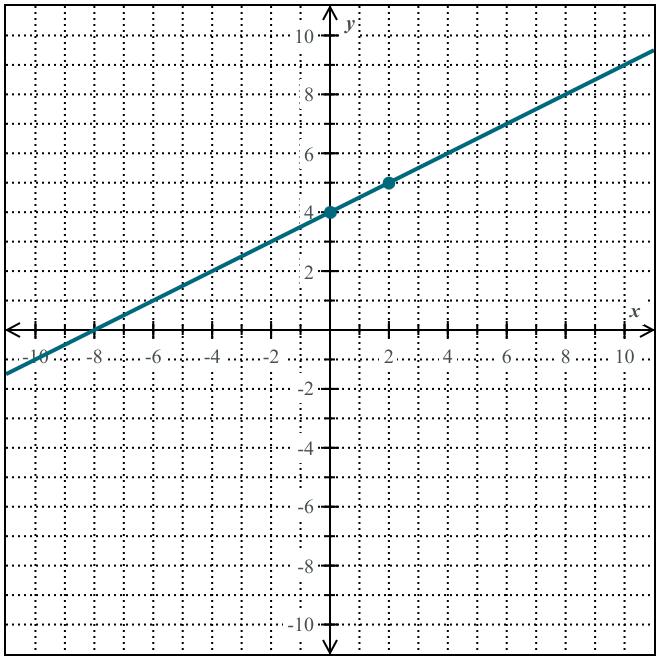
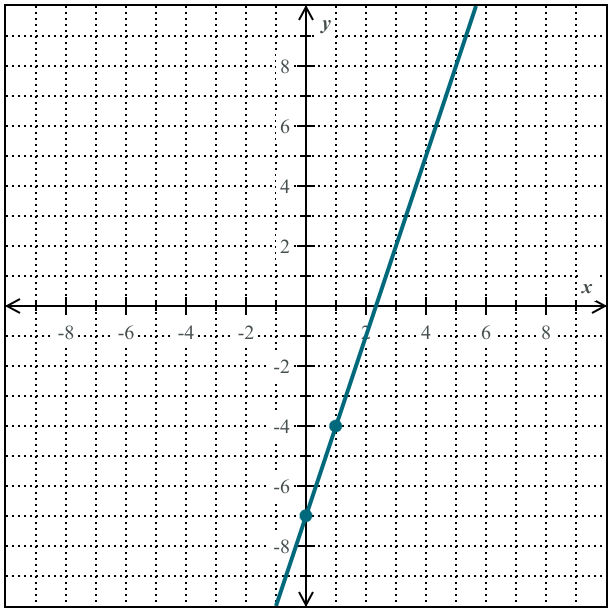
Question 8 of 38

slope: $\frac{1}{2}$

y-intercept: 4



Question 9 of 38



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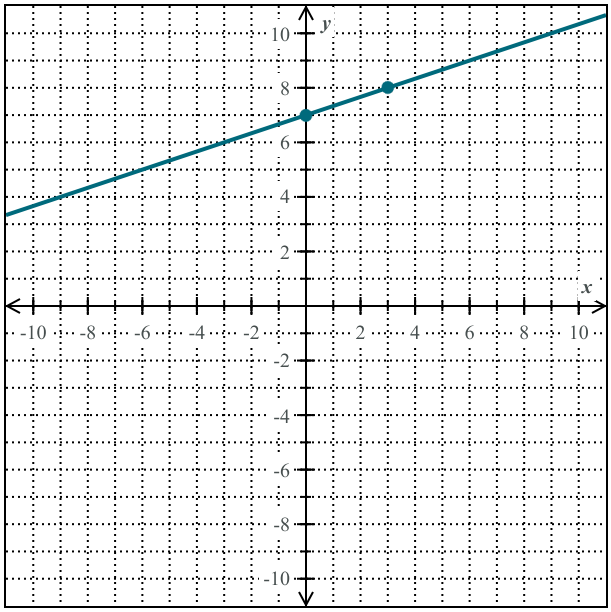
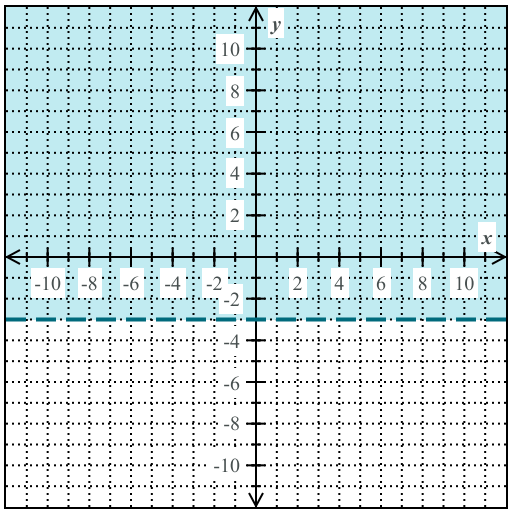


Question 11 of 38

(x, y)	Is it a solution?	
	Yes	**No**
$(-6, 5)$	◉	○
$(0, -2)$	◉	○
$(4, -5)$	○	◉
$(-8, -3)$	○	◉

Question 12 of 38



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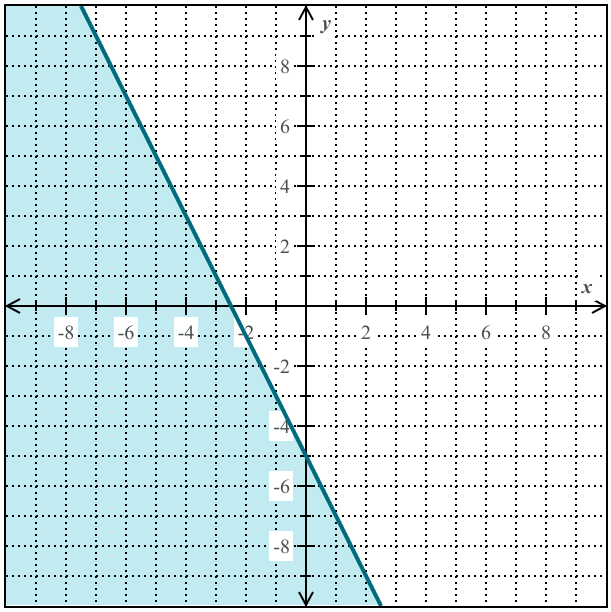

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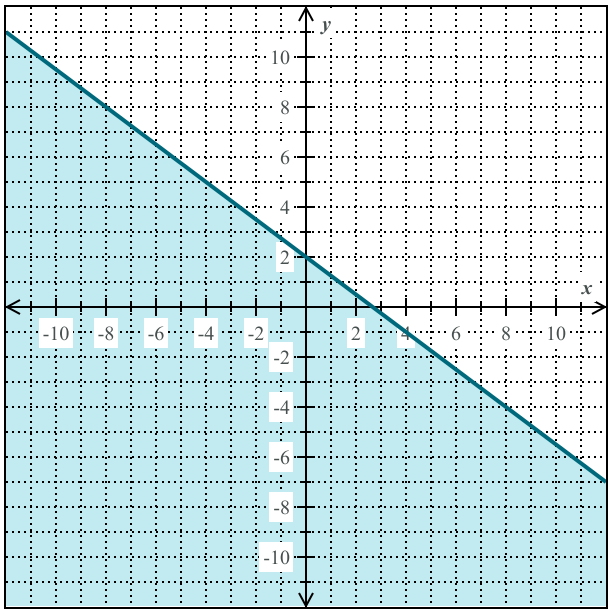

Question 15 of 38

$35n + 135 \leq p$

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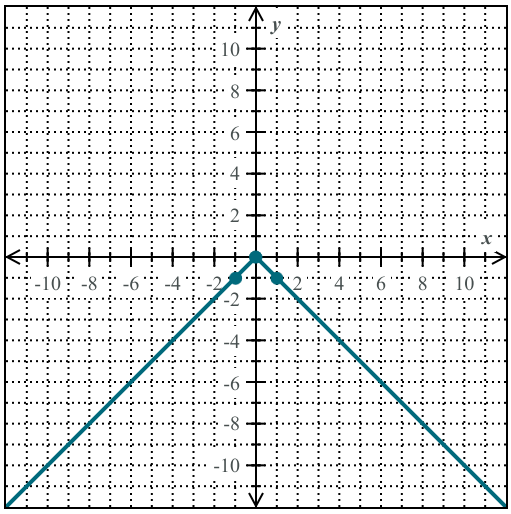
$$f(-2) = -3$$

$$f(1) = -1$$

$$f(3) = -2$$

Question 17 of 38



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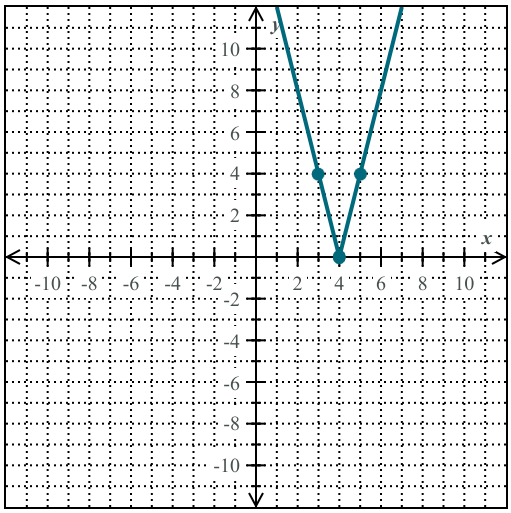

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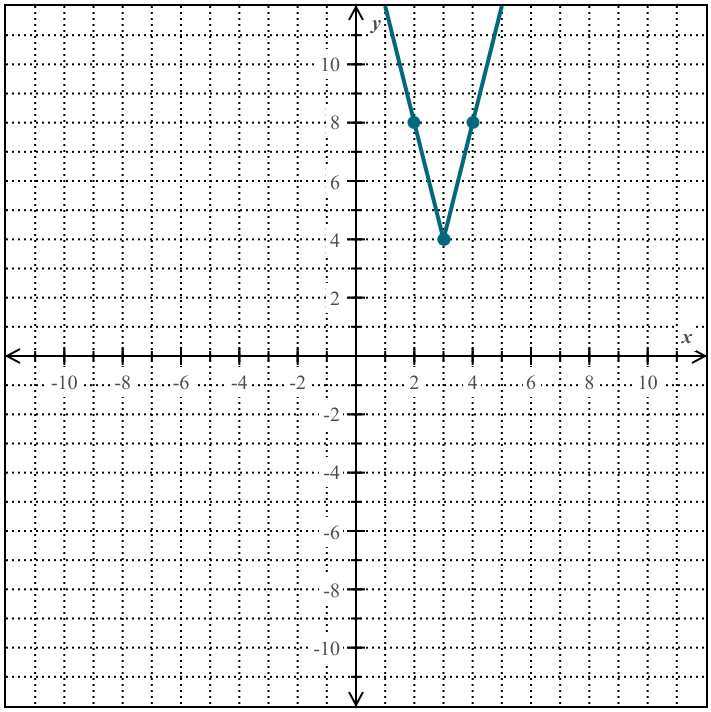

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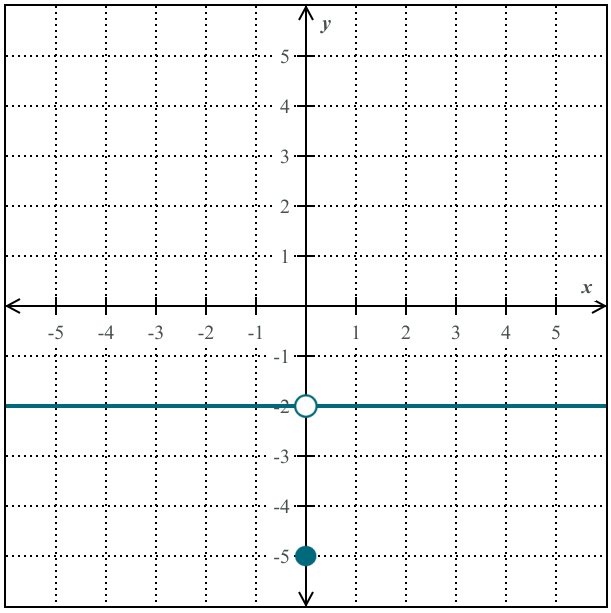

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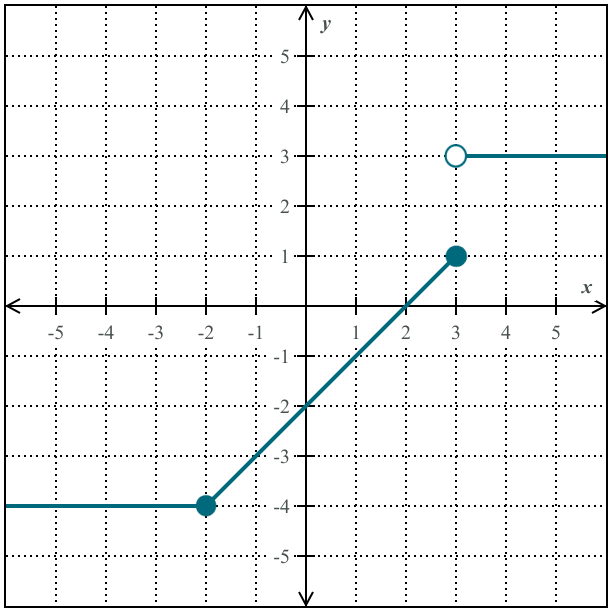

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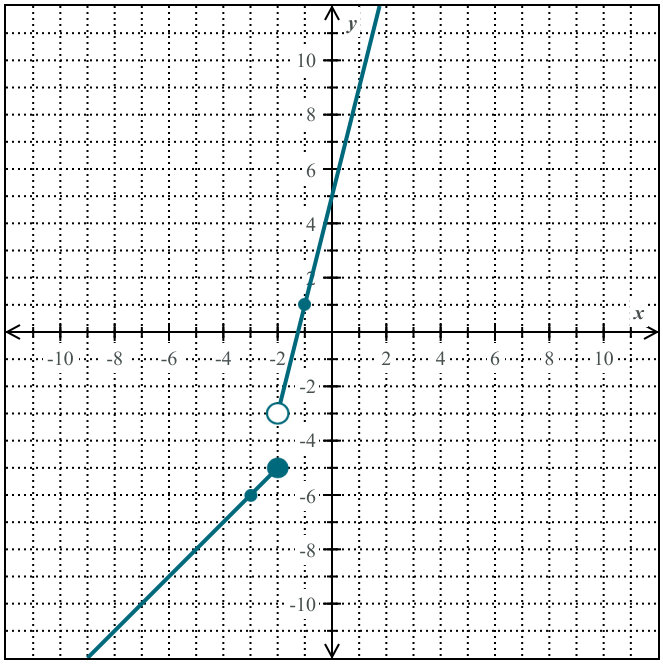

Is the function continuous?

- Yes
- No

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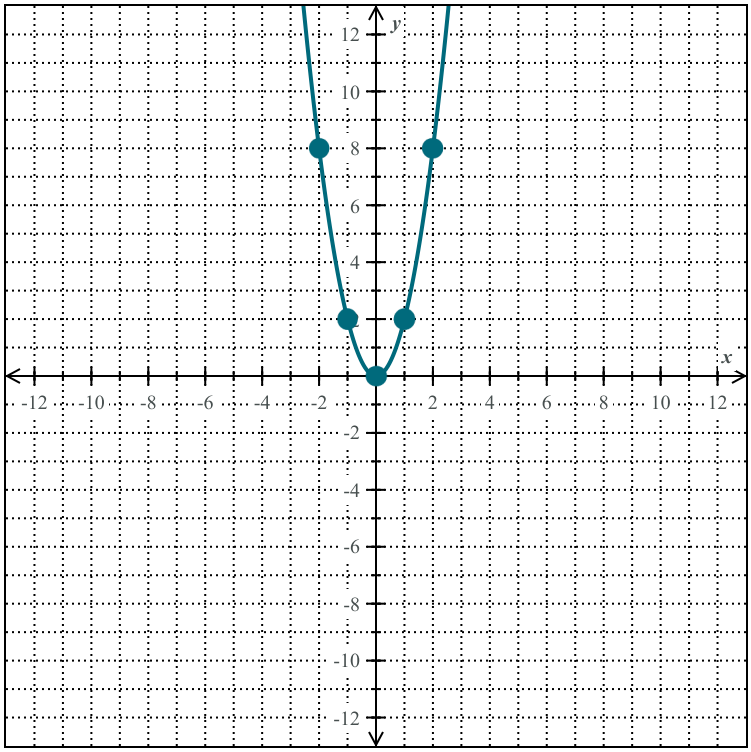

Note that the graph of the function $f(x) = 2x^2$ is a parabola. Its vertex is $(0,\ 0)$.

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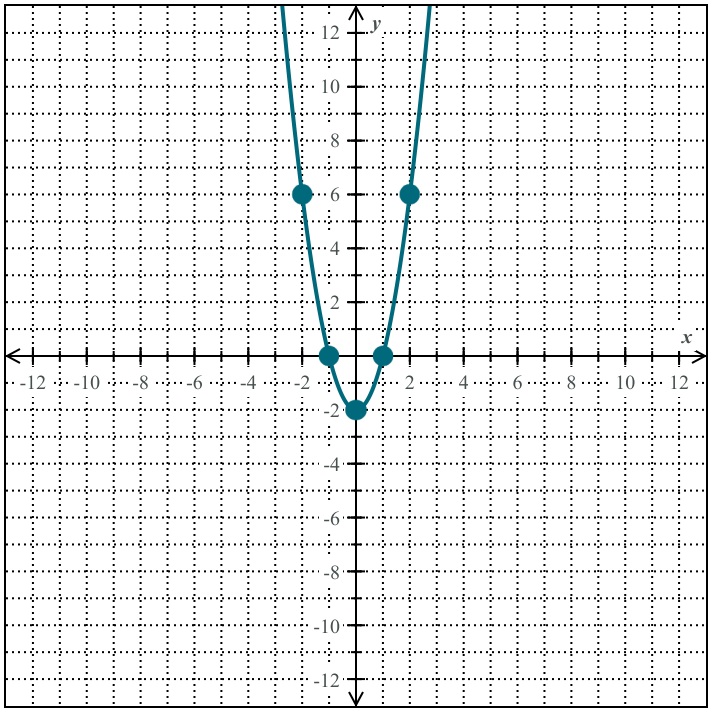

Note that the graph of the function $f(x) = 2x^2 - 2$ is a parabola. Its vertex is $(0, -2)$.

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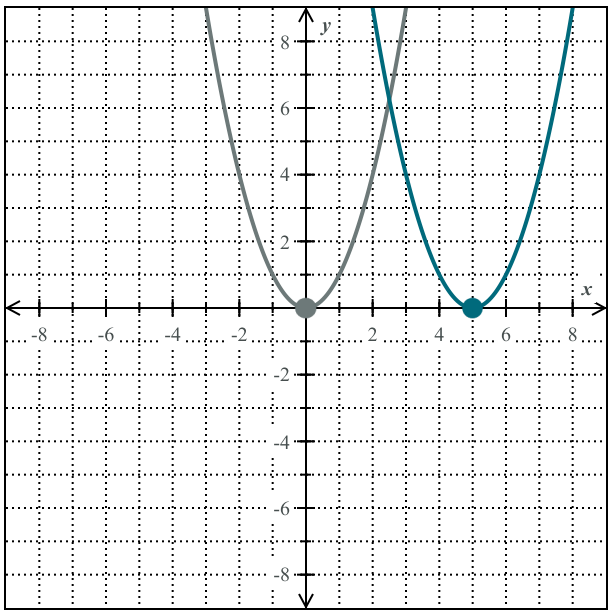
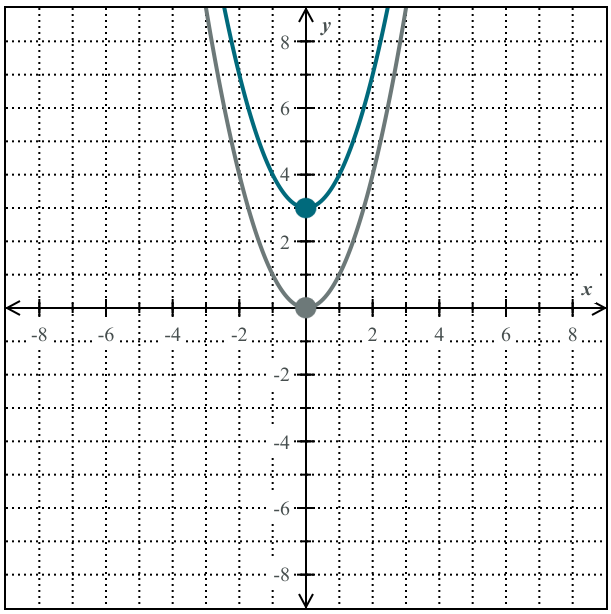
(a)



(b)



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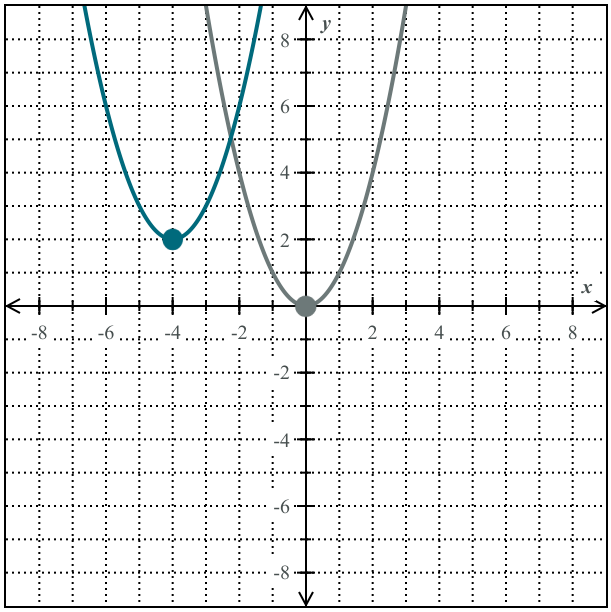

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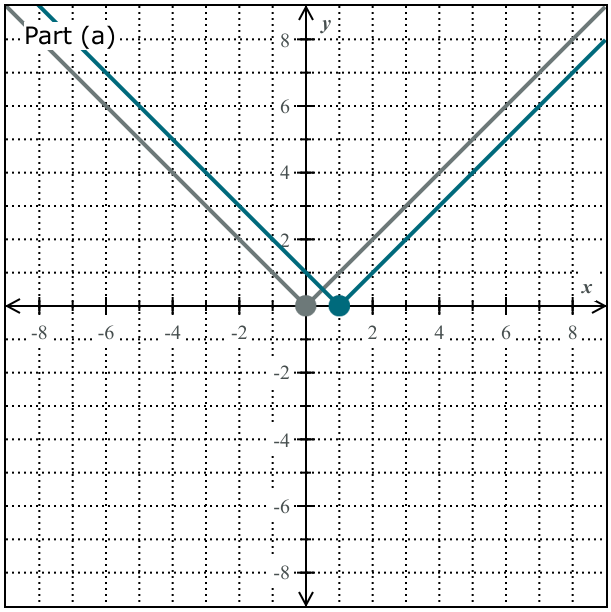

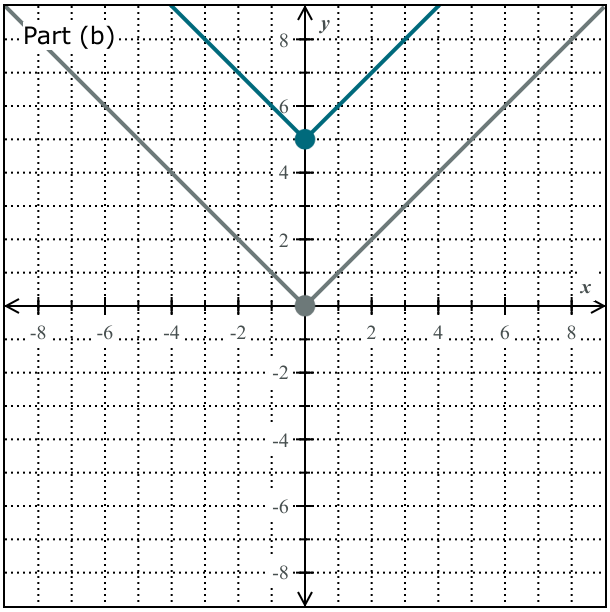

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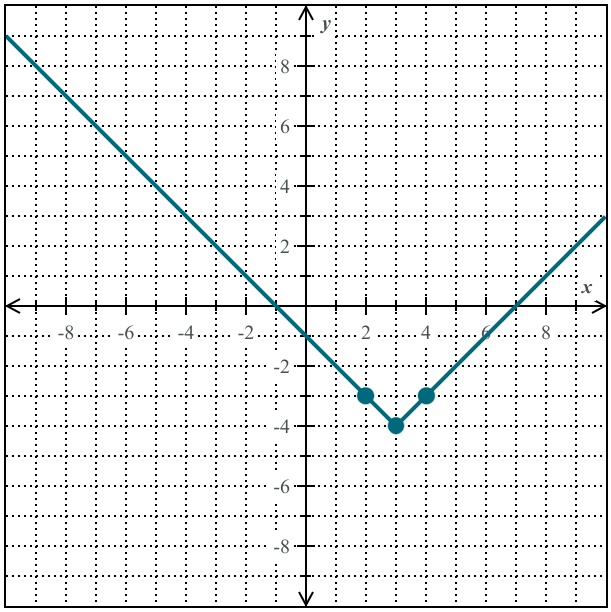

Question 30 of 38

$f(x) = 2x^2 - 2$

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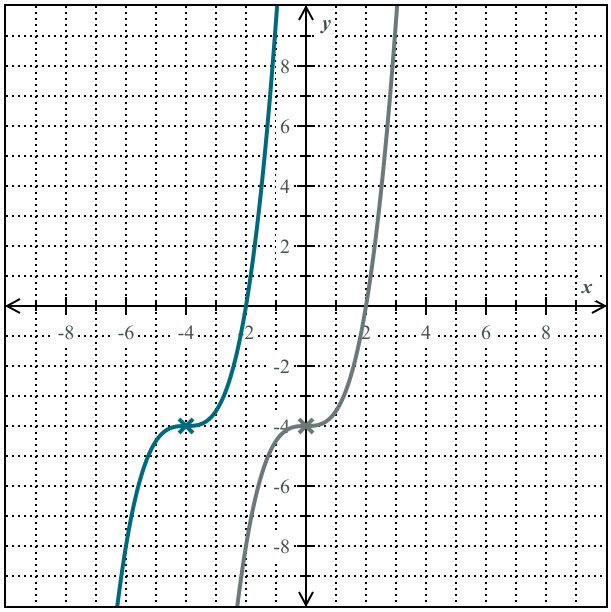
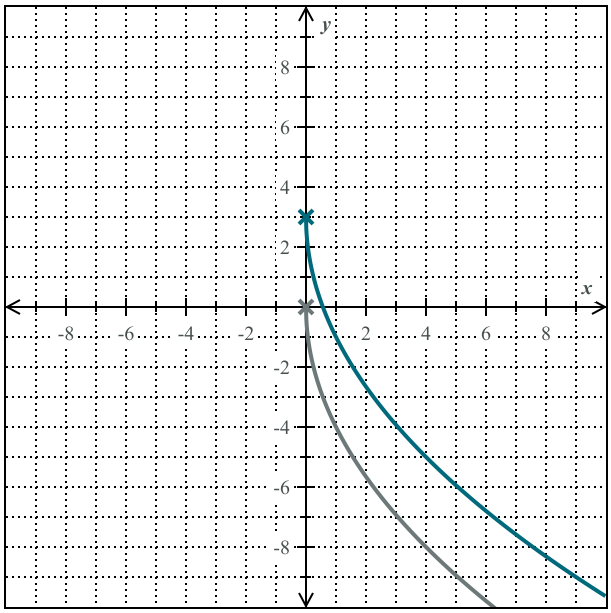
(a) The graph of $y=f(x)$ is shown. Translate it to get the graph of $y=f(x+4)$.



(b) The graph of $y=g(x)$ is shown. Translate it to get the graph of $y=g(x)+3$.



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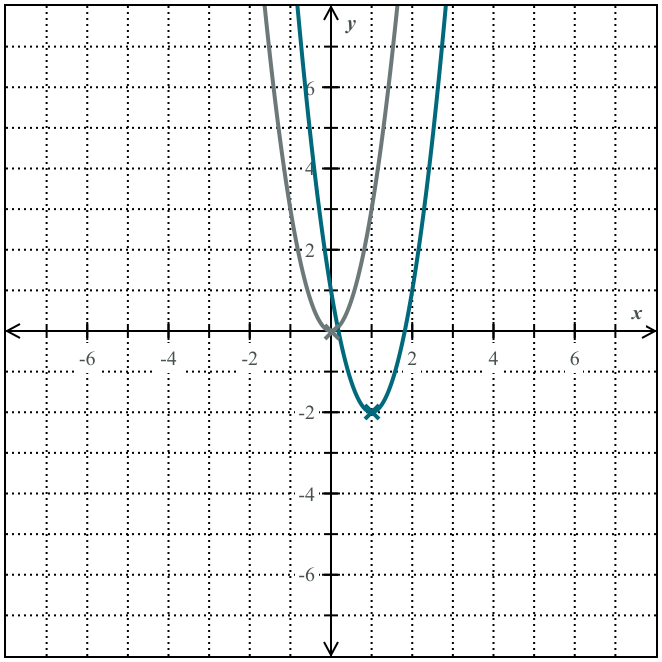

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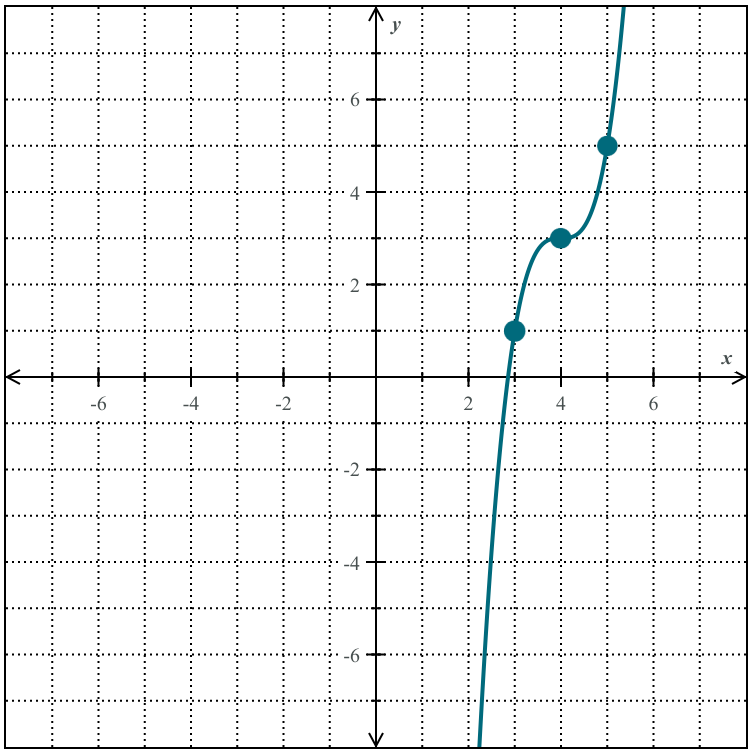

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$$h(x) = -\sqrt{x-2} - 3$$

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y-intercept: -1

slope: $\frac{5}{3}$

Question 36 of 38

(a) range: $y \geq 0$

(b) domain: All reals

Question 37 of 38

$g(x) = |x-4| - 2$

Question 38 of 38

Function 1

$\{(5, 1), (6, 5), (7, 8), (8, 9)\}$

- ○ Linear
- ◉ Not linear

Function 2

$\{(1, 0), (4, -2), (7, -4), (10, -2)\}$

- ○ Linear
- ◉ Not linear

Function 3

x	y
-4	-5
0	-7
4	-9
8	-11

- ◉ Linear
- ○ Not linear

Function 4

x	y
-3	3
-1	9
1	27
3	81

- ○ Linear
- ◉ Not linear

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